TULA: A Risk-Controlled Safety Layer for Automatic Billing

Toll Uncertainty and Liability Assessment

Suryansh Pratap Singh · Working paper, version 2, 25 October 2026

> **Status.** Working draft. The risk controller, evidence check, human review budget, learning-from-disputes loop and benchmark simulator are built and tested. One reader (open-source plate OCR) has been measured on real Indian plates. The Muse Spark reader is built but has not been run yet, and SAM 3.1 is not built yet. Results in Section 7 come from **simulated traffic** unless marked "real photos"; they test the method, not a real toll operator. Every hypothesis is stated so that it can fail.

Abstract

Barrier-free tolling bills a vehicle with no person checking the bill. A camera reads the plate, a reader picks up the tag, and a fee follows. When the plate is misread, or the tag belongs to another car, a real owner gets a penalty. Today's systems guard against this with one confidence threshold. That threshold cannot say how often a wrong charge will happen, it says nothing about night, rain or an attacker with a cloned plate, and it gives no safe way to improve the system after deployment.

TULA is a safety layer that sits between the camera and the bill. It fuses independent evidence into one disagreement score, then decides to **charge**, **hold**, or **send to a human**. A conformal risk controller sets the charge cut-off so that the wrong-charge rate stays below a stated target, calibrated separately for each group of traffic (plate style, region, day or night, weather). A group too small to certify is never charged automatically. TULA also answers two questions a deployed system faces: **how strict a promise can it make for a given number of human reviewers**, and **how can it keep learning from its own decisions** when the only feedback is disputes from the people it charged. We show, in simulation, that a system which learns from disputes slides to about 2.5 times its promised wrong-charge rate while believing it is on target, and that a small random audit keeps the promise. The design is not specific to India: the same layer applies to any automatic penalty (tolls, e-challans, fraud flags), in any country.

The name is Sanskrit. *Tulā* (तुला) is the balance scale. The system weighs the evidence before it punishes anyone.

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1. Problem

Free-flow tolling is used in Ireland, Norway, Chile, Australia, Canada, the Czech Republic and several US states, and India started Multi-Lane Free Flow (MLFF) tolling in May 2026, with the first plaza on NH-48 and bids out for 104 more. In India a misread becomes an e-notice, and the fee doubles after 72 hours.

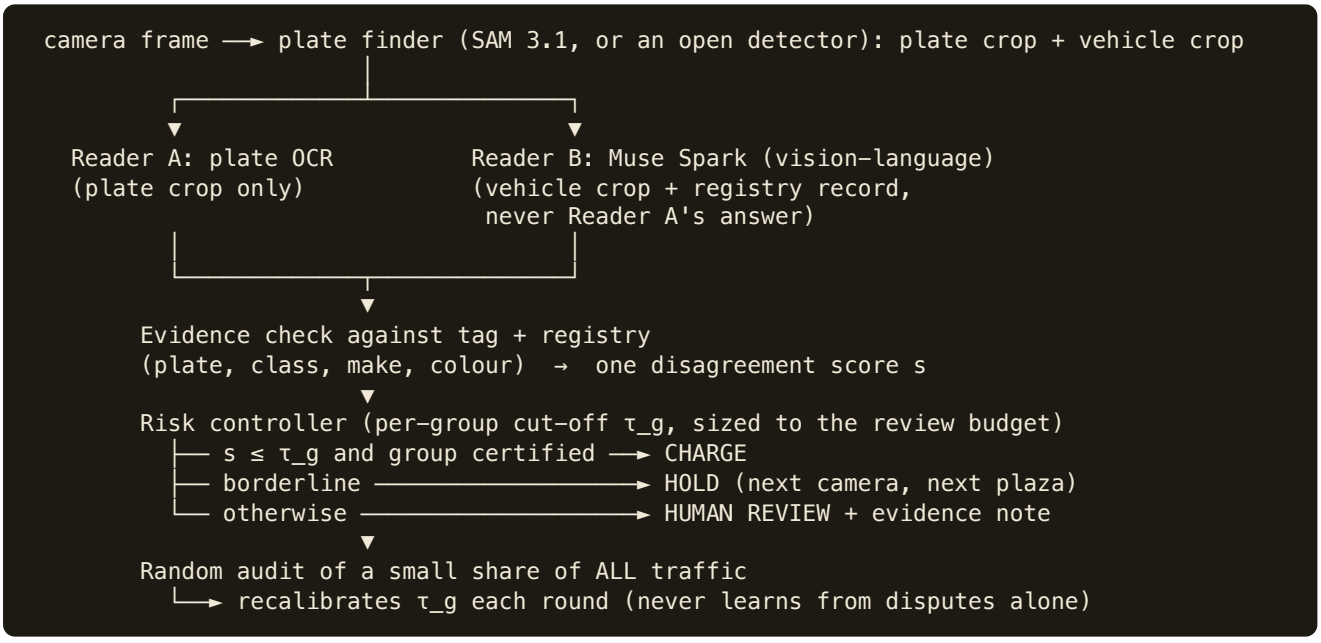
The errors are small in rate and large in count:

- India refunded about 12.55 lakh faulty FASTag deductions in 2024, roughly 0.03% of transactions, and that was with toll booths still in place.
- A New York State audit found about 36,000 wrong charges at one Thruway exit in three weeks, roughly a 5% error rate.
- Dublin's M50 logged 14,280 plate misreads in about 87 million passages.

Four things are missing from current systems:

1. **No stated error rate.** A confidence threshold has no known wrong-charge rate, per plate type or per region.
2. **No attack accounting.** Printed stickers, cloned plates and swapped tags are documented. No deployed system reports how much damage they can do.
3. **No safe default for rare groups.** A model that is accurate on average can be wrong far more often on hand-painted plates, two-line plates, fog or night.
4. **No safe way to learn after deployment.** The only feedback is disputes, and only people who were charged can dispute. Learning from that feedback teaches the system that its unnoticed mistakes were correct.

2. The idea in one picture



3. Method

3.1 Readers that cannot copy each other

Two readers are only useful if their mistakes differ. If the second reader sees the first reader's answer, it tends to agree with it and the disagreement signal disappears. TULA applies the need-to-know rule from our earlier work on agent context (ZeroBleed): each reader receives only the inputs it needs, and the evidence check is ordinary code that sees both outputs.

- **Reader A** is a small specialist OCR model (today: the open-source fast-plate-ocr model `cct-s-v2-global`). It sees the plate crop and nothing else.

- **Reader B** is Muse Spark through the Meta Model API. It reads the plate again and returns plate text, confidence, plate style, state, vehicle type, make, colour and image problems as structured JSON. It is shown the registry record and asked whether the image is consistent with it. It is never shown Reader A's answer.

Language models invent plausible plates. Asking Reader B to check against a fixed registry record, and logging when it cannot, keeps an invented plate from becoming evidence.

3.2 The disagreement score

The evidence check decides who would be billed (the tag's account, or the registry match for the plate when there is no tag) and turns the evidence into signals: plate disagreement between the readers, mismatch between the tag's registered plate and the read, vehicle type against the billed toll class, colour and make against the registry, low reader confidence, and a missing tag. These are combined into one number $s \in [0, 1]$, higher meaning more doubt.

Two scores are supported: a hand-set weighted sum, and a logistic regression fitted on a separate split of the data. Fitting the score is allowed because the cut-off is calibrated on different passages afterwards. The safety claim comes from calibration, not from the score: a better score gives more automation, a worse score gives more holds, and neither breaks the bound.

3.3 The risk controller

Define the loss for one passage and a cut-off τ:

$$L(\tau) = 1[\text{ vehicle is charged } \text{ AND } \text{ charge is wrong }] \qquad \text{charged means } s \le \tau$$

"Wrong" means the bill goes to the wrong account or for the wrong amount. L is bounded by 1 and never decreases as τ grows. This is the setting of conformal risk control (Angelopoulos et al., 2022). With n calibration passages and empirical risk $\hat{R}_n(\tau)$, choose

$$\hat{t} = \sup \{ \tau : \ n/(n+1) \cdot \hat{R}_n(\tau) + 1/(n+1) \ \le \ \alpha \}$$

Then, if test passages are exchangeable with the calibration passages, $E[L(\hat{t})] \le \alpha$. The target α is the wrong-charge rate we promise.

That bound holds **on average over calibration sets**. A single deployment wants more: the promise should hold for *this* calibration set with high probability. TULA therefore also offers a high-probability mode in the style of Learn-then-Test: it picks the largest τ whose one-sided Clopper-Pearson upper bound on the wrong-charge rate stays below α at confidence $1 - \delta$ (we use $\delta = 0.1$). Both modes are implemented and tested; the tests check that the average risk stays under α in the first mode and that the bound is exceeded in at most a δ share of trials in the second.

3.4 Per-group cut-offs, and when to refuse

The bound above holds over all traffic. A group that is rare or hard can carry far more than α while the average still looks fine. TULA calibrates one cut-off $\hat{t}_g$ for each group g (plate style, region, day or night, weather), so the bound holds inside each group.

This has a hard floor. Because of the `1/(n+1)` term, a group with `n_g` calibration passages can only certify `α` if `n_g ≥ 1/α − 1`. For `α = 10⁻⁴` that is about **10,000 passages per group**.

So TULA adopts a rule we call **no certificate, no automatic charge**: a group without enough calibration data to certify `α` is never charged automatically. Its passages are held or reviewed, and the group is reported as uncertified. In our simulation the uncertified groups were exactly the hardest ones (hand-painted plates in fog or at night, two-line plates at night in fog): the system gets less automatic where it knows least.

To reach realistic error rates without waiting for real wrong charges, the benchmark injects errors (tag swaps, cloned plates, misreads) at higher-than-real rates. Re-weighting them back to realistic frequencies is planned, not built.

3.5 What an attacker can take away

Conformal guarantees assume exchangeability. A cloned plate or a sticker attack breaks it on purpose. We do not claim the guarantee survives an attack unchanged. We state what it becomes.

Proposition. Let `P` be the calibration distribution and `Q` the test distribution, and let `TV(P, Q) = ε`. For any loss bounded in `[0, 1]`, `E_Q[L(t̂)] ≤ E_P[L(t̂)] + ε ≤ α + ε`.

Proof sketch. For a function `f` with values in `[0, 1]`, `|E_Q f − E_P f| ≤ TV(P, Q)`. Apply it to `f = L(t̂)` with `t̂` fixed after calibration. ∎

If an attacker controls a fraction `ε` of the traffic, the wrong-charge rate is at most `α + ε`. The bound is loose, but it is a number we can publish and test.

3.6 The human review budget

Every hold or review costs something. An operator has a fixed number of reviewers, so the real question is: **with reviewers for a share `b` of traffic, what is the strictest promise `α` TULA can make?** TULA sweeps `α`, measures the share of traffic sent to humans at each level, and reports for each budget the smallest `α` whose review load fits. If the review queue still overflows (for example during a surge), passages are triaged by expected harm, doubt times fee. The highest-harm passages go to humans first, the rest wait on hold, and nothing over budget is ever charged. This turns the three-way trade-off between wrong charges, automation and human work into one table an operator can read (Section 7.3).

3.7 Learning from its own decisions safely (the recursive loop)

After deployment a billing system receives feedback only through disputes, and disputes are **selective labels**: only charged people can dispute, and many wrongly charged people never do. A cloned-plate victim may not notice a small toll at all. A system that recalibrates on this feedback treats "no dispute" as "correct". Its estimated error falls, its cut-off loosens, it charges more, and the loop repeats. Attackers add pressure: when cloned plates get through, cloning spreads.

TULA never recalibrates on disputes alone. Each round, a small **uniform random audit** of all traffic (charged, held and reviewed alike) is checked by humans. Those audited passages are a random sample of real traffic, so they are exchangeable with future traffic in a stationary period, and the bound of Section 3.3 still applies to cut-offs calibrated on them. The audit also gives an unbiased estimate of the live wrong-charge rate, so the system knows its real number instead of the dispute-based one. Disputes are still handled as complaints and refunds. They are simply not used as calibration labels.

The limit is drift. Audits from older rounds describe older traffic, so when the attack mix changes quickly the bound can lag by a round. Section 7.4 shows this happening once, and the audit catching it.

3.8 Explanations that serve a real decision

For every hold and review, Muse Spark writes a short note from the structured evidence, for example: "Plate reads match, but the vehicle type and colour do not match the registered vehicle." The note goes to the reviewer and, if the driver disputes, to the driver.

4. What is new, and what is not

To our knowledge, based on 15 targeted searches on 24 October and a re-check of every cited paper on 25 October, the building blocks exist separately and the combination does not. We have not found a paper, patent or product that bounds the wrong-charge rate in automatic billing, sizes that bound to a human review budget, or studies safe recalibration from dispute feedback in this setting. This is a first pass and we will keep trying to break the claim.

Piece	Closest existing work	Status in TULA
Confidence thresholds before manual review	USPTO 10019640, 8768753	Replaced by a calibrated cut-off with a stated rate
Uncertainty in plate reading	Benchmarking probabilistic LPR (2023)	Reused as an input signal, not claimed
VLMs reading plates	LP-LLM; BLPR (confidence-driven VLM fallback)	Reused; checked against a registry record
Clone-plate detection by make/model vs registry	UK False Plate Classifier; make-model classifiers (arXiv 1809.00953)	Not claimed as new
Attacks on plate readers	GRAPHITE; street-legal adversarial plate rim (SPAR)	Used as the attack set
Conformal risk control	Angelopoulos et al., 2022	The core tool, not claimed
Conditional / adaptive risk control	Automatically Adaptive CRC (arXiv 2406.17819)	Related; TULA uses explicit groups and a refusal rule
Robust conformal prediction	Gendler et al. ICLR 2022; Lip-RCP (arXiv 2506.05434)	Pixel-noise robustness; TULA's bound is about attacker-controlled traffic
Deferral to humans under workload limits	DeCCaF (arXiv 2403.06906); learning to defer	Related; TULA ties the deferral budget to a certified error bound
AI control with an audit budget	Ctrl-Z (arXiv 2504.10374)	Evaluation style borrowed for billing
Selective labels from human decisions	Lakkaraju et al., 2017	The problem behind Section 3.7
Reader independence by input isolation	none found for plate reading	Our design choice
No-certificate-no-charge rule for small groups	none found	Our design choice
Review-budget frontier for a certified billing bound	none found	New in TULA

Piece	Closest existing work	Status in TULA
Random-audit recalibration vs dispute-label recalibration	none found for billing	New in TULA
Open benchmark pairing tag, plate, class and billing	none found	Released with the project

The safest wording is "to our knowledge, the first", and only after we repeat the search before release.

5. Threat model

- **Cloned plate.** The right plate on the wrong vehicle. Both readers can agree, so the vehicle check against the registry carries the decision.
- **Swapped tag.** A tag moved to a different car. The plate-vs-tag and class checks should catch it.
- **Wrong class.** A tag registered under a cheaper toll class. The vehicle-type check should catch it.
- **Sticker or edited image.** Printed patterns that mislead plate readers. Plan: a transfer attack on Reader A first, then an adaptive attack on the full pipeline.
- **Adaptive attackers over time.** Attackers copy what works. The recursive-loop experiment lets cloning grow with its success.
- **Out of scope for now.** Collusion between reviewers, tampering with the registry itself, and attacks on the camera hardware.

6. Evaluation

Data. Public data only, from more than one source, so the method is not tied to one country's plates.

Layer	Source	Status
Indian plates with plate text	Hugging Face `zenitsu09/indian-number-plate` (1,684 unique photos, 965 plates; licence not stated)	Downloaded, used for Reader A
Foggy and hazy Indian plates	Foggy-Hazy License Plates, Mendeley (1,001 photos with clear versions, CC BY 4.0)	Downloaded; plate text being hand-labelled
Plate locations	Kaggle "Indian License Plates with Labels" (2,083 photos, boxes only)	Downloaded
A second country	Public plate set from outside India	To choose
Tag, registry, billing	Simulated, with tag misreads, swaps, clones and wrong classes	Built

The Indian Licence Plate Dataset in the wild (16,192 images), named in version 1, is not public: its authors write that it cannot be released "because of legalities involved in making Indian Road data public". IEEE DataPort's Indian LPR set needs a paid subscription.

Baselines. Today's biller (trust the tag, else the plate), one confidence threshold, the fused score with a fixed cut-off, OCR alone with the controller, Muse Spark alone with the controller.

Metrics. Wrong-charge rate (the promised number), automation rate, hold rate, human review load, worst-group rate, groups over the target, the believed-vs-true gap in the recursive loop, and how much of the guarantee survives each attack.

Hypotheses, each of which can fail.

1. The controller keeps the wrong-charge rate at or below α in every certified group, while the single-threshold baseline exceeds α in at least one group.
2. The fused system gives a higher automation rate than OCR alone at the same wrong-charge rate.
3. Isolating the readers' inputs lowers the correlation of their errors compared with showing Reader B the OCR answer.
4. Under a cloned-plate attack on a fraction ε of traffic, the measured wrong-charge rate stays under $\alpha + \varepsilon$.
5. Calibrating on day and testing on night breaks the guarantee, and the group-wise controller recovers it only when the group is calibrated.
6. A system that recalibrates on dispute labels exceeds its promise and under-estimates its own error; random-audit recalibration keeps the promise.
7. Reader confidence alone is not a safe charging signal on real Indian plates: some reads the model is very sure of are wrong.

7. Early results

All numbers below are from one seed. Readers in 7.1, 7.3 and 7.4 are **simulated** with assumed error rates, so these results test the method, not real plates.

7.1 The controller on simulated traffic (hypotheses 1 and 2)

120,000 passages over 20,000 vehicles, split into fit, calibration and test; 24 groups; target $\alpha = 0.5\%$ per group. Planted problems (tag swaps, clones, wrong classes, missing tags) are about 10% of traffic, far above real life.

Method	Wrong-charge rate	Charged automatically	Worst group	Groups over α
Today's biller (trust the tag)	6.19%	100%	7.39%	24/24
One confidence threshold (0.9)	4.23%	67.6%	5.48%	12/24
Fused score, fixed cut-off	1.30%	90.5%	2.05%	15/24
OCR only + controller	0.44%	29.3%	0.90%	7/24
Muse only + controller	0.47%	89.2%	1.14%	6/24
TULA (fused + controller)	0.46%	89.4%	1.01%	7/24
TULA, 90%-confidence mode	**0.30%**	**83.5%**	**0.58%**	**1/24**

Hypothesis 1 holds in the high-probability mode. In the average-case mode, single test splits still land over α in several groups, which is what an in-expectation bound allows. Hypothesis 2 is mixed: fusion beats OCR alone by far (89% vs 29% automation), but in simulation it is no better than Reader B alone, because the simulated Reader B already carries the vehicle checks. Real readers will decide this.

7.2 Reader A on real photos (hypothesis 7)

fast-plate-ocr `cct-s-v2-global` on 1,684 labelled Indian plate photos: **31% exact reads**, 22.5% character error rate (49% exact on crops 200 px or wider, 13% on crops under 100 px). Of the 313 reads it was at least 90% sure of, **69 were wrong (22%)**. A billing system that trusts that confidence would charge the

wrong person on those plates. This supports hypothesis 7 and is the direct case for calibrating on outcomes instead of trusting confidence.

7.3 The human review budget (Section 3.6)

Reviewers can check	Strictest promise that fits	Wrong charges (test)	Charged automatically
2% of traffic	1.00%	0.79%	88.7%
5% of traffic	0.75%	0.59%	87.6%
10% of traffic	0.30%	0.18%	74.1%
20% of traffic	0.20%	0.10%	50.8%

Stricter promises cost both reviewers and automation, and the table makes that price explicit.

7.4 Learning from disputes (Section 3.7, hypothesis 6)

Promise `α = 1%`, six rounds of 30,000 passages, wrong charges disputed with probability 0.3 (clones) to 0.7 (wrong class); cloning grows with its success against the naive system.

Round	Clone rate	Naive: true	Naive: believes	TULA: true	TULA: believes
1	2.0%	0.46%	0.24%	0.46%	0.55%
2	2.5%	2.26%	1.18%	0.47%	0.54%
3	6.3%	2.27%	1.13%	0.68%	0.69%
4	11.3%	2.57%	1.10%	0.85%	0.67%
5	15.0%	2.80%	1.12%	1.14%	1.73%
6	15.0%	2.42%	1.05%	0.89%	0.77%

The naive system settles at 2.3–2.8%, more than twice its promise, while believing it is near 1.1%. TULA, recalibrating on a 3% random audit, stays near its promise and knows its real number. In round 5, when cloning jumped fastest, TULA slipped to 1.14%; its own audit flagged the rise (estimated 1.73%), and it was back under 1% in round 6. That is the drift limit of Section 3.7, measured.

8. Limits and ethics

- **The guarantee is conditional.** It holds when calibration traffic looks like real traffic. Sections 3.5 and 3.7 and hypothesis 5 are there to show how it fails.
- **Simulated parts are simulated.** The tag, registry and billing layers, the simulated readers' error rates and the dispute probabilities are our assumptions. Results on them measure the method, not an operator's real error rates.
- **Rare groups cost review time.** No certificate, no automatic charge means more human work where data is thin. That is a deliberate trade.
- **A human stays in the loop.** TULA reduces wrong charges and never removes the reviewer.
- **Privacy.** Public datasets only. Nothing in the repository identifies a vehicle owner, and raw datasets are not redistributed.
- **Reuse beyond tolling.** The same layer applies to any automatic penalty (e-challans, fraud flags) in any country. Those are future work, and the tolling results should not be assumed to carry over.

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